AACQUILINE LINU VARGHESE

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Summary

B.Tech Computer Science & Engineering graduate specializing in AI & Deep Learning, with hands-on experience in full-stack development, computer vision pipelines, and production-grade AI system integration via APIs.

Education

LBS Institute of Technology for Women(LBSITW) *2022 – 2026*
Computer Science & Engineering CGPA: 8.53/10
Relevant Coursework: Data Structures, Algorithms, AI/ML, Deep Learning, Computer Vision, Database Management, Operating Systems, Computer Networking

Government Higher Secondary School *2019 – 2021*
Computer Science Percentage: 95.67/100

Skills

Programming Languages: Python, Java, C, JavaScript, HTML, CSS, SQL
AI & ML: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib
Web Development: Node.js, Express, React.js, REST APIs
DevOps & Tools: Git/GitHub, VS Code, Postman, Jupyter Notebook, Google Colab
Databases: PostgreSQL, Oracle DB, MongoDB
Cloud & APIs: Google Gemini Flash API, Gmail OAuth 2.0, Google Calendar API, Judge0, Twilio REST API

Experience

National Institute of Technology, Calicut, IEEE Summer Intern *May 2025 – June 2025*

- Engineered a document image denoising system using Spatial Attention U-Net and Autoencoders in PyTorch, improving visual clarity and achieving a PSNR increase from 7.63 dB to 16.95 dB through optimal performance tuning.
- Processed and augmented a dataset of 216 image pairs, designing full training/evaluation pipelines using L1 Loss and Adam Optimizer to focus on critical text regions via a Spatial Attention Module (SAM)
- Collaborated with internal stakeholders to solve problems and validate model performance against industry-standard benchmarks, ensuring optimized output for downstream OCR tasks employing project management methodologies for better results.

LBS Centre for Science and Technology, Machine Learning Intern *March 2025 – April 2025*

- Built and compared software solutions encompassing statistical modeling, classification and regression models using Decision Trees, KNN, and Logistic Regression across multiple datasets, debugging code across Python environments.
- Applied Scikit-learn and Pandas to perform data cleaning, data quality checks and data visualization, strengthening model evaluation through comparative metric analysis

Projects

Placera: Placement Assistance & Support System | Node.js, React.js, Docker, JWT, PostgreSQL *Nov 2025 – April 2026*

- Architected a multi-channel notification engine delivering real-time placement alerts via Web Push (VAPID/Service Worker), Gmail OAuth using Nodemailer with OAuth 2.0 refresh token flow, and Twilio WhatsApp REST API, with node-cron scheduled reminders, ensuring timely placement updates across 1,000+ students.
- Engineered an ATS Resume Checker pipeline supporting PDF, DOCX, and OCR-scanned documents using pdf-parse, mammoth, and tesseract.js, with a weighted scoring model (Skills 50%, Profile 30%, CGPA 20%) delivering real-time compatibility feedback and actionable skill gap analysis.
- Built a gamified DSA coding practice microservice integrating a self-hosted Judge0 Community Edition instance for sandboxed multi-language code execution (Python, Java, C++, JavaScript, C) with Gemini AI code review, a three-tier hint system, XP/badge gamification, and a college-wide leaderboard backed by MongoDB.
- Developed and integrated the Admin Dashboard, Job Management, Eligibility Verification Engine, Training & Attendance Management, Placement Statistics & Analytics, Application Tracking, Alumni Portal, and AI-based Placement Level Scoring modules using Node.js, Express, React.js, and PostgreSQL across 25 relational tables.

Crowd Monitoring and Alert System | YOLOv11m, OpenCV, Flask, Twilio *Oct 2024 – Mar 2025*

- Built a real-time density analyzer using YOLOv11m and Gaussian kernel density mapping, performing feature engineering to achieve an average processing rate of 27-32 FPS on standard GPUs
- Developed an automated safety protocol using Twilio API (WhatsApp/Email) to trigger alerts when density exceeded the 4.0 people/m^2 safety threshold.
- Implemented Lucas-Kanade Optical Flow and NMS in OpenCV to track movement speed and filter redundant person detections in high-density scenarios ensuring the robustness of the interface
- Designed a real-time dashboard using Python (Flask) and MongoDB to provide data visualization for heatmaps, people counts and historical density trends.

Certifications & Achievements

- **NPTEL**: Cloud Computing (IIT Kharagpur), **IBM SkillsBuild**: Data Analytics, **Google**: Cybersecurity, **Udemy**: Full Stack Web Development
- **Best Project Award** : Arduino Simulation (IEEE RAS SBC LBSITW)